

Anthropomorphism in Chatbot Systems Between Gender and Individual Differences

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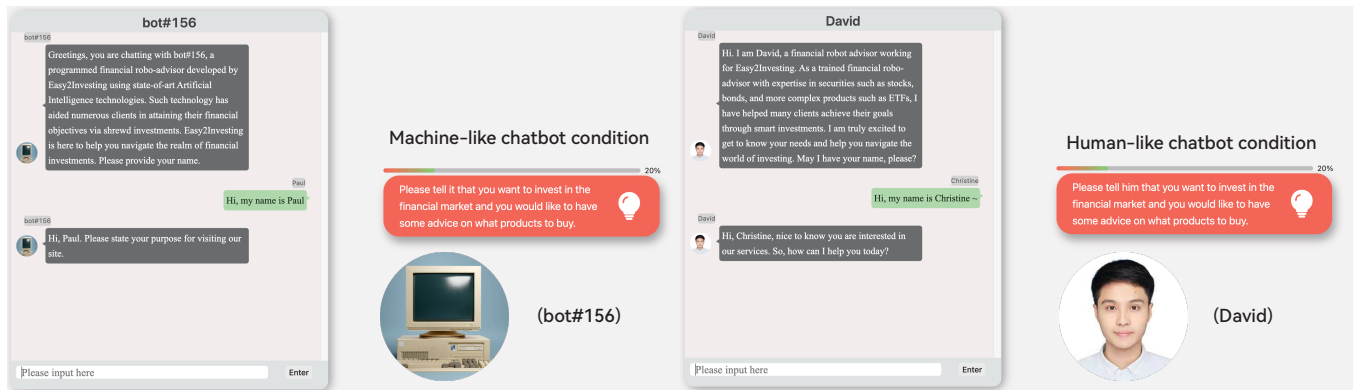


Figure 1: Two chatbot designs were presented: machine-like (on the left) and human-like (on the right).

ABSTRACT

Online advisory chatbots are becoming increasingly popular and are often used to support user decision-making in different sectors. The design of such systems has been the object of investigation since it can alter the user's perception and cognition. This study investigates the effect of chatbot design, gender, and individual tendencies on perceived anthropomorphism in a financial investment scenario. Our online experiment with 116 participants revealed that manipulating chatbot design can alter the user perception of anthropomorphism contingent on both gender and anthropomorphism tendencies. Male users with a strong anthropomorphism tendency exhibited a higher degree of perceived anthropomorphism compared to female users with a similar tendency. However, this effect was only observed when the chatbots themselves lacked design elements that promote anthropomorphism. Our novel finding provides theoretical and practical implications to the domain of chatbot system design and applications.

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CCS CONCEPTS

• **Human-centered computing** → **User studies**; *Web-based interaction*; Empirical studies in HCI.

KEYWORDS

Anthropomorphism, Chatbot Design, Gender, Online Experiment

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1 INTRODUCTION

Anthropomorphism is the tendency to attribute human traits and feelings to non-human entities, including artificial intelligence (AI) [1, 10, 28]. Design choices deliberately harness this tendency, as evidenced by the human-like features of robots and the conversational skills of chatbots [4]. These anthropomorphic AI systems are increasingly permeating our lives. For example, digital assistants whisper from our phones, chatbots handle customer service across industries, and even complex tasks are tackled by tools like ChatGPT [6, 12, 16]. As this trend continues, these technologies seem destined to become woven into the fabric of every human activity, even influencing our decisions.

While anthropomorphizing AI can enhance user engagement and adoption [5, 25], it is worth noting that the perceived level of

“humanness” in AI varies greatly among individuals. Factors like personality, gender, and cultural background all play a role [7, 21, 27]. This raises concerns that the benefits of designing for anthropomorphism might only reach a specific subset of users, potentially excluding others and even perpetuating biases.

In this research, we designed a chatbot system resembling a financial advisor and conducted an online experiment to study this phenomenon. We found that the anthropomorphic design influences how people perceive the chatbot’s human-like qualities through a questionnaire evaluation with 116 participants. The analysis results revealed that male users with a high tendency to anthropomorphize tend to perceive chatbots as more human-like than female users with the same tendency. However, this only happens when chatbots are not designed with anthropomorphic features. The paper is structured as follows. The next section gives the background of the study, and then we explain the hypothesis development. After that, we show the method, analysis, and result. Finally, we conclude this work by discussing the significance of these outcomes.

2 BACKGROUND

When it comes to anthropomorphism in AI, two key questions emerge: how can we design it, and how do users experience it? The first stream dives into design strategies that intentionally nudge users into perceiving AI as more human-like. The second investigates the individual differences that shape how users perceive and interact with AI.

2.1 Anthropomorphic Design

The spontaneous attribution of human characteristics to non-human entities appears to be an inherent feature of human cognition [1, 10, 28]. The line between human and machine blurs even further when our natural anthropomorphizing tendency meets the scientific community’s push for “designed anthropomorphism” [22]. Meanwhile, social norms and dynamics guide the communication between humans, and they also apply to human-computer interaction. Social response theory entails that people naturally apply these norms when interacting with computers, and they do so even when they are fully aware that such machines do not have human feelings or intentions [17, 18]. This innate tendency to connect with technology has driven the design of robots and AI systems to increasingly resemble humans, leveraging our natural social instincts to enhance user experience and interaction [9, 15].

Beyond crafting human-like appearances, researchers strive to endow machines with human-like behaviors and minds [4]. Several design cues for conversational agents to resemble human communication have been proposed. Research in this field identifies three basic categories of design cues: identity design, verbal design, and non-verbal design cues [24]. **Identity design cues** refer to those elements that emphasize human identity and are not inherently present in machines (e.g., gender). Seeger and colleagues [24] also included visual cues in this category. However, visual cues can be considered a separate category on its own given that some non-visual information (e.g., the name of the chatbot is Tom) can independently signal gender (i.e., identity information) without displaying any visual object – e.g., a picture of a man [2], and that

some anthropomorphic designs (e.g., adding human eyes to a machine) are purely visual and do not add any explicit or implicit identity information [14]. **Verbal design cues** include the use of self-reference pronouns such as “I” and the use of emotional expressions that are often used in human-to-human conversations, such as the expression “Well done!” [24]. **Non-verbal design cues** refer to those subtle behaviors that, in human-to-human communication, give conversational rhythm and a sense of timing – e.g., turn-taking, pausing when the other is speaking, pausing while thinking, etc. [24]. In physical robots, several behavioral cues can be manipulated to enhance anthropomorphism (e.g., body language, facial expressions, etc.). However, in virtual artifacts (e.g., chatbot), there are obvious limitations. In such a case, verbal and non-verbal cues, which henceforth will be grouped into a single category called communication cues, are the main cues that confer anthropomorphic behavior to chatbots. In addition, it is worth noting that using either verbal or non-verbal cues in isolation can be counterproductive. For instance, the research found that using only non-verbal cues decreases anthropomorphism; however, this effect reverses when non-verbal cues are added [24].

To sum up, the amplification of anthropomorphism in virtual AI (e.g., chatbot systems) can be achieved by using visual, identity, and communication cues. In addition, using all three cues together synergistically (not in isolation) increases the anthropomorphic perception of AI artifacts [2].

2.2 Anthropomorphism Tendency and Gender

Our interactions with the world are colored by our inherent tendency to see humanness in everything, even non-human objects [2]. However, this “humanness radar” varies in intensity from person to person, and this variation is influenced by factors like personality, background, and past experiences [10]. Importantly, the strength of our individual anthropomorphism tendency affects how much trust, empathy, and concern we extend to non-human agents like robots or AI [10]. This has crucial implications for the design of anthropomorphic technologies used in high-risk situations. Recognizing this variability, Waytz et al. developed the Individual Differences in Anthropomorphism Questionnaire (IDAQ), a reliable tool for measuring individual anthropomorphism tendency and widely used in research [27]. Despite its importance, individual differences are often overlooked in design practices, possibly due to the cost of acquiring such information [4]. As a result, technology is often designed for the “average user”, which might be sufficient for low-risk settings. However, with the rise of AI and its increasing integration into sensitive areas like finance and healthcare [20], where mistakes can have significant consequences, factoring in individual variations in anthropomorphism becomes crucial. Studies exploring the influence of gender on anthropomorphism add a further layer of complexity. Early research suggests that men seem to anthropomorphize robots more than women [23]. Similarly, children exhibit higher anthropomorphism towards non-human objects whose gender matches their own [21]. Interestingly, adult male users also appear to prefer anthropomorphic robots over their female counterparts [7]. In summary, anthropomorphism is an innate tendency that has been enhanced, in recent years, by certain design choices

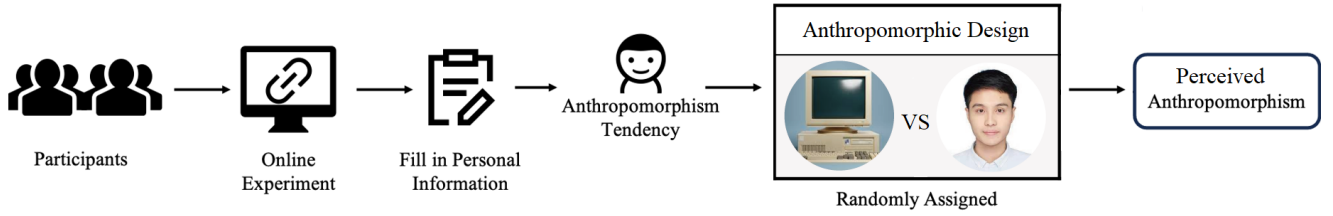


Figure 2: The research pipeline: (1) MTurk recruitment, (2) online platform redirection, (3) demographic data collection, (4) anthropomorphism tendency pre-test, (5) randomized chatbot interaction (machine-like or human-like) in a financial simulation, and (6) post-test measurement of perceived anthropomorphism.

in technological artifacts (e.g., AI) and seems to vary across individuals depending on the user’s anthropomorphism tendency and gender.

3 HYPOTHESES DEVELOPMENT

Our research concentrates on AI solutions, specifically advisory chatbots, which are rapidly gaining traction in web and mobile apps. This study aims to uncover the extent to which anthropomorphic design influences the perceived anthropomorphism of chatbots for individuals with diverse gender identities and anthropomorphism tendencies.

To successfully anthropomorphize the digital beings, visual, identity, and communicative cues are essential [2, 24]. Prior research has established that anthropomorphic design, in general, increases perceived anthropomorphism in chatbots [2]. Building upon this body of research, our hypothesis serves a confirmatory purpose, evaluating our manipulation (i.e., manipulation check) and laying the groundwork for subsequent hypotheses:

- **H1:** Perceived anthropomorphism of a chatbot will be significantly enhanced when its design incorporates human-like characteristics, as opposed to designs without such elements.

The variety of research on anthropomorphism reveals potential influence by gender, with hints that men might be more inclined to project human qualities onto AI [23]. However, this perspective often overshadows the diverse landscapes of individual predisposition towards anthropomorphization, shaped by innate characteristics and nurtured experiences. In our study, we unravel this intricate interplay between gender and personal inclinations towards anthropomorphism. Consequently, our second hypothesis unveils that:

- **H2:** Anthropomorphism tendency interacts with gender on perceived anthropomorphism. Specifically, men exhibit a higher

perceived anthropomorphism than women only when the tendency for anthropomorphism is high.

Finally, we posit that the effect described in **H2** is dependent on the chatbot’s design. Specifically, this gender gap might only emerge when the chatbot presents a machine-like appearance. When the chatbot exhibits human-like features, a greater level of perceived anthropomorphism might be evident for all users, regardless of gender. Therefore, our next hypothesis posits that:

- **H3:** Anthropomorphism tendency interacts with gender and with the chatbot’s design on perceived anthropomorphism. Specifically, only when the chatbot is not anthropomorphized does a difference between male and female users with a high anthropomorphism tendency occur.

4 METHOD

Our investigation unfolded through a web-based chatbot application. Inspired by the scenario utilized by Bao et al. [2] in their study, which itself stemmed from Holzmeister et al.’s work [13], participants embarked on a simulated investment scenario, seeking advice from the chatbot advisor online. Figure 2 visually summarizes the research pipeline. The study protocol was approved by the Research Ethics Committee of The Chinese University of Hong Kong, Shenzhen.

4.1 Stimuli

Our investigation required two distinct chatbot personas, crafted to test the impact of design on perceived anthropomorphism. Employing the stimuli designed by previous research [2, 24], we created a machine-like and a human-like chatbot condition. To achieve this differentiation, we carefully tweaked visual, identity (*bot#156* vs. *David*), and communication cues (see Figure 1).

Table 1: The table, with references, describes the variables and questionnaires used in the experiment.

Variable	Construct measured	Reference	Use
Anthropomorphic Design	Cues Manipulation	[2]	IV
Gender	Male vs. Female	-	IV
IDAQ	Anthropomorphism Tendency	[27]	Covariate
Godspeed	Perceived Anthropomorphism	[3]	DV

4.2 Variables and Experimental Design

The experiment's dependent variable (DV) was **Perceived Anthropomorphism**, measured using the Godspeed scale [3]. Our study's independent variable (IV) was **Anthropomorphic Design**, with two conditions: machine-like vs. human-like. We ran a between-subject online experiment where participants were randomly assigned. We also collected participants' gender (male, female, and other) and anthropomorphism tendency (i.e., the tendency users have to anthropomorphize non-human entities) using the IDAQ scale developed by Waytz et al. [27]. For visualization purposes, this latter variable was dichotomized into low, medium, and high using the first and third quantiles. A summary of the variables and questionnaires used in the study, with related references, can be found in Table 1.

4.3 Procedure and Participants

The procedure unfolded within the framework established by Bao et al. [2], ensuring consistency and reliable comparison. Participants were first redirected to our online platform, where they received pre-experiment information on the study's purpose and data privacy. After consenting to the experiment's terms and conditions, participants were asked to provide basic demographic information, including age, gender, etc. After that, the IDAQ measured participants' tendency for anthropomorphism. Then, participants were randomly assigned to either the machine-like or human-like chatbot condition. Following the instructions displayed in a dedicated box on the left of the screen (see Figure 1), participants engaged in a simulated investment consultation with the chatbot to obtain financial advice on investing their savings. The chatbot was designed to incrementally acquire the details in the order of income, investing goals and risk tolerance. After that, the chatbot, armed with this personal snapshot, crafted bespoke investment advice for each participant. To foster open communication and minimize biases, before the consultation started, participants were assured that the advisory service belonged to a neutral non-profit organization, free from any vested interests in any specific recommendations. Finally, after concluding the conversation, participants answered questions on the Godspeed scale to assess their perceived anthropomorphism of the chatbot.

To unravel the underlying interactions with the chatbot, we recruited 116 participants. Our participant pool comprised 56 students from a local international university and 60 individuals through Amazon Mechanical Turk. This resulted in a balanced group of 59 females and 57 males, ranging in age from 18 to 71 years old, with an average of 30.53 ($SD = 14.48$) in female and 32.02 ($SD = 11.55$) in male. Ensuring clear communication and familiarity with the digital landscape, we recruited only participants who were fluent in English and experienced with using online chatbot services.

4.4 Results

Our gender composition revealed a binary landscape, with no participants selecting "other" as their identity. As detailed in Table 2, we analyzed the combined effect of chatbot design and participant gender on the perceived anthropomorphism of the chatbot, measured using the IDAQ scale. The results revealed that the human-like chatbot elicited a stronger sense of anthropomorphic response

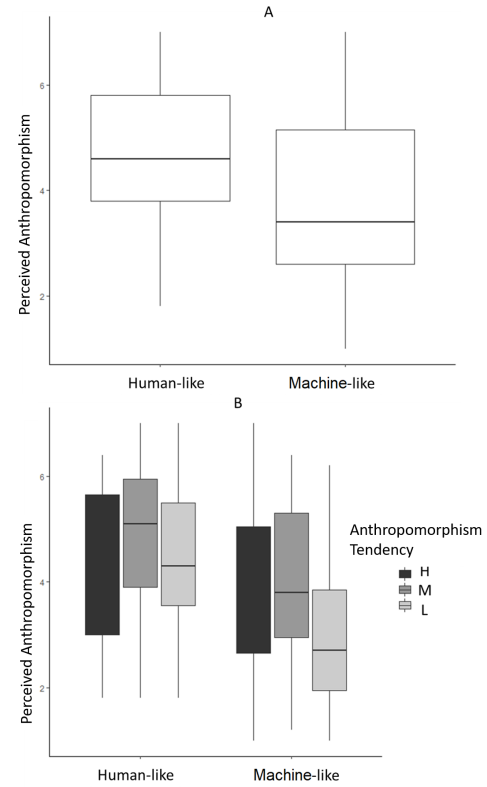


Figure 3: Box-plots comparing (A) perceived anthropomorphism under conditions by (B) different levels (high, medium, and low) of anthropomorphism tendency.

compared to the machine-like design, with average IDAQ scores of 4.59 ($SD = 1.46$) and 3.68 ($SD = 1.58$), respectively. However, when we explored the gender dimension, the scales of perceived anthropomorphism remained remarkably balanced, with male (IDAQ $mean = 4.09$, $SD = 1.66$) and female participants (IDAQ $mean = 4.15$, $SD = 1.52$) exhibiting no significant difference in their perception of the chatbots.

The box-plot in Figure 3A presents a comparison of perceived anthropomorphism across the two chatbot design conditions. As expected, the human-like design elicited higher IDAQ scores than

Table 2: The statistical results of the two different conditions of the anthropomorphic manipulation and gender for perceived anthropomorphism.

Variable	Condition	Mean	SD
Anthropomorphic Design	Human-like chatbot	4.59	1.46
	Machine-like chatbot	3.68	1.58
Gender	Male	4.09	1.66
	Female	4.15	1.52

the machine-like design, indicating a stronger perception of anthropomorphism. Figure 3B further explores this interaction by looking at different levels of participant anthropomorphism tendency (high, medium, and low), represented by varying shades of gray. The separated groups consisted of 30 (Age *mean* = 24.73, *SD* = 6.18), 54 (Age *mean* = 30.87, *SD* = 11.80) and 32 (Age *mean* = 38.00, *SD* = 16.61) participants respectively. Interestingly, this figure reveals that participants with medium anthropomorphism tendencies seem to exhibit the highest overall perceived anthropomorphism, compared to both high and low-tendency groups.

To test our hypotheses, we run a three-way ANOVA. From the analysis, it can be noted that manipulating anthropomorphism via design cues had the intended effect $F(1, 104) = 13.01, p < .001$, Partial $\eta^2 = .11$. This supported our first hypothesis **H1**. The effect of the co-variate anthropomorphism tendency is also significant $F(2, 104) = 3.18, p = .046$, Partial $\eta^2 = .06$. This is expected since, in general, people with a higher tendency to anthropomorphize objects will tend to be higher in perceived anthropomorphism [27]. The effect of gender was not significant $F(1, 104) = 0.13, p = .7$, Partial $\eta^2 = .002$. To further explore the interaction between design and individual tendencies, we conducted a post-hoc analysis with Tukey's Honest Significant Difference test. The pairwise comparison of the three levels of anthropomorphism tendency shows a significant difference between low and medium anthropomorphism tendency $p = .05$ with a moderate effect size (Cohen's $d = 0.52$). However, comparisons between low and high ($p = .82$) and medium and high ($p = .23$) tendencies were not significant.

The analysis also shows two significant interaction effects. The first involved a two-way interaction between anthropomorphism tendency and participant gender $F(2, 104) = 6.12, p = .003$, Partial $\eta^2 = .11$, revealing a significant influence on perceived chatbot anthropomorphism. This suggests that how participants naturally perceive human-like qualities in objects might interact with their gender in shaping their perception of the chatbot. Meanwhile, a three-way interaction between anthropomorphism tendency, gender, and anthropomorphic design cues significantly impacts perceptions of the chatbot's human likeness: $F(2, 104) = 3.98, p = .021$, Partial $\eta^2 = .07$. This hints that the intricate interplay of individual tendencies, gender, and the chatbot's visual cues creates a unique symphony of perception.

To delve deeper into the interplay between variables and examine **H2** and **H3**, we resort to three interaction plots presented in Figure 4. The first plot (Figure 4A) explores how gender and individual anthropomorphism tendencies interact to influence perceived anthropomorphism in the entire dataset. The remaining plots (Figure 4, B and C) dissect this relationship separately for the machine-like and human-like chatbot conditions. Figure 4A reveals that a high anthropomorphism tendency amplifies perceived anthropomorphism for male participants but dampens it for female participants. Conversely, medium to low tendencies seem to have the opposite effect, with females perceiving higher human-likeness and males perceiving lower. This supports **H2**, suggesting that only individuals with high anthropomorphism tendencies exhibit a gender difference in perceived anthropomorphism, with males perceiving the chatbot as more human-like than females.

When looking at the machine-like condition (Figure 4B), we find a similar gender effect but with an even stronger interaction,

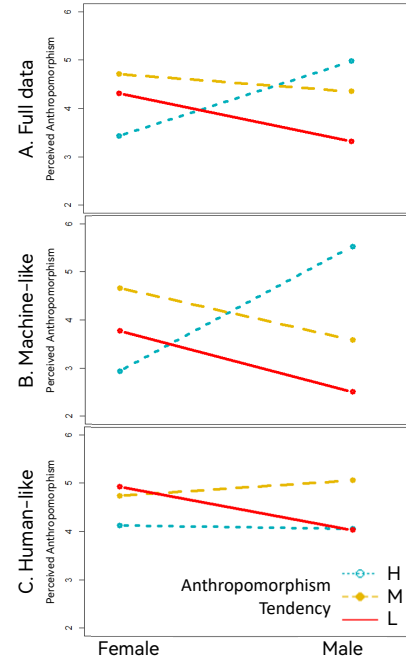


Figure 4: Interaction plots showing how gender interacts with anthropomorphism tendency for the (A) full dataset and by different conditions, (B) machine-like and (C) human-like.

particularly for participants with high anthropomorphism tendencies. This further bolsters **H2**, suggesting that the gender difference holds even when the chatbot lacks overt human-like cues. However, the human-like condition (Figure 4C) reveals no such disparity. Aligned with **H3**, it appears that only when the chatbot is not anthropomorphized do men with high anthropomorphism tendencies demonstrate a higher perceived human-likeness compared to women with similar tendencies. This effect disappears when the chatbot adopts human-like cues, as both high-tendency males and females exhibit similar responses. The plots also indicate that, regardless of design, female participants with low anthropomorphism tendencies seem to report higher perceived human-likeness compared to males.

5 DISCUSSION

Drawing on a financial scenario established in prior research, we examine how a chatbot's design, a user's gender, and their inherent anthropomorphism tendency influence the perception of the chatbot's human-like qualities in this research. Prior work has emphasized the critical role of chatbot design in shaping perceived anthropomorphism [2, 4, 24], highlighting its connection to various measures of technology effectiveness [9, 15]. Our findings not only reaffirm this importance but also extend our understanding by identifying individual factors that moderate this effect. While user gender alone did not directly influence perceived anthropomorphism, we uncovered an interplay with individual tendencies to anthropomorphize. Specifically, we observed male users with a higher tendency to anthropomorphize objects tend to display a stronger anthropomorphic perception compared to female users

with similar tendencies. This effect, however, seems to manifest when the chatbot lacks overt human cues primarily. When the chatbot adopts a human-like appearance and behavior, the gender difference among high-tendency individuals disappears. Additionally, for users with a low tendency to anthropomorphize, our data suggests that females, regardless of design, might perceive the chatbot as slightly more human-like than males.

Implication. Previous research has pointed out that an anthropomorphic response is more likely to be found when relevant anthropocentric knowledge is available, potentially via experience, when there is an increase in motivation for social engagement, and when there is social isolation [8]. These factors might indirectly relate to individual differences in anthropomorphism tendency. Research on gender differences shows that, on average, males tend to be more engaged with playing video games, using computers, and with technology in general [11, 19], and tend to be more socially isolated than females [26]. This suggests that when a chatbot adopts strong anthropomorphic design cues, an overall positive response might be observed regardless of the user's gender. However, when the chatbot displays less evident human-like features, individual differences in prior technology experience and gender-specific tendencies related to social engagement might play a more significant role in shaping anthropomorphic responses.

Our findings offer insights for AI designers, urging them to consider the nuanced interplay between users' gender and chatbot design, particularly when aiming for high levels of perceived anthropomorphism. For male users with a strong natural tendency to anthropomorphize, highly human-like chatbot designs might paradoxically lead to a lower perception of anthropomorphism. Conversely, for female users, such designs could be strategically employed to amplify the intended impression of anthropomorphism, aligning with their potentially higher receptiveness to human-like cues. In essence, tailoring design choices to individual differences in gender and anthropomorphism tendency can optimize the effectiveness of highly anthropomorphic AI systems.

Limitation. While our investigation focused primarily on gender and individual variations in anthropomorphism tendency, it's crucial to acknowledge that other individual and situational factors may also play a role in shaping perceived anthropomorphism in AI systems. Previous research highlights the influence of social isolation tendencies and personal experience on this perception [8]. This suggests that personality traits and exposure to technology deserve further exploration in future research. To address this limitation, our next step is to delve into personality-related factors and prior experience with computers and technology to discover how they interact with individual tendencies and gender to influence perceived anthropomorphism. Such comprehensive insights could potentially extend our findings and shed light on the mechanisms behind the observed gender differences in our study.

6 CONCLUSION

This research investigates the multifaceted determinants of perceived anthropomorphism in chatbot interactions. We dissect the roles of chatbot design, user gender, and individual anthropomorphism tendency through the development and deployment of a

real-world chatbot system in online experiments. Our findings illuminate the intricate interplay between these factors, revealing implications that transcend the immediate experimental context. In light of the rapid proliferation of chatbots, these insights contribute to chatbot design and online services literature, offering both theoretical and practical insights for crafting human-centered chatbot experiences.

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